

Towards a statistical theory of optimal quantization

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Abstract. In this paper we suggest an approach to unify some concepts of quantization of empirical data along the principles of statistical decision theory. In this way various quantization methods can be evaluated in decision theoretic terms, and particular quantization methods can be distinguished as being optimal for statistical decision problems. The approach of this paper is strongly inspired by ideas of Bock (1996). The basic concept is the relation of majorization of measures. The main conclusions follow from the theory of comparison of experiments by Blackwell (1951, 1953) and from mathematical results by Pötzlberger (2000). For algorithmic aspects of our approach we refer to Pötzlberger and Strasser (2000).

1 Introduction

Roughly speaking, the concept of quantization can be described as follows: A distribution σ on a Borelfield $\mathcal{F}$ of a vector space V is to be replaced by a distribution τ with finite support. The substitution is to be carried out in such a way that the data governed by σ are summarized and the complexity of the distribution is reduced.

Let us explain the formal concept of a quantization by the special case where the starting point is a partition of the data. This construction will be basic for our subsequent considerations. Moreover, it illustrates the strong relation between cluster analysis and quantization.

Let $\mathcal{B} = (B_1, B_2, \dots, B_m) \subseteq \mathcal{F}$ be a partition of V and let $\sigma(B_i) > 0$ for every $i = 1, 2, \dots, m$. If we denote

$$\sigma(A|B_i) := \frac{\sigma(A \cap B_i)}{\sigma(B_i)}, A \in \mathcal{F},$$

then the distribution σ can be written as

$$\sigma = \sum_{i=1}^m \sigma(B_i) \sigma(\cdot|B_i),$$

i.e. as a mixture of distributions with disjoint supports. Now, we define a measure τ with finite support in the following way: Let the mass points of the measure τ be the centroids

$$t_i := \int x \sigma(dx|B_i)$$

of the mixture components and define the weights of the measure τ to be $w_i := \sigma(B_i)$. Then we have

$$\tau := \sum_{i=1}^m w_i \delta_{t_i},$$

where δ_x denotes the one-point measure at x . The measure τ will then be called the quantization of σ defined by the partition $\mathcal{B}$, and it will be denoted by the symbol $\tau = \sigma^{\mathcal{B}}$.

For practical purposes it would be sufficient to consider only quantizations which are defined by partitions in the way just described. However, it is possible to extend this construction in such a way that the general concept of majorization of measures may be applied. The historical starting point of majorization was measuring the concentration of wealth in a human population. Details on the general theory of majorization can be found in Marshall and Olkin (1979). The connection between quantization and majorization will turn out to be very fruitful for a theoretical analysis of quantization.

We want to quantize a distribution σ in such a way that the loss of information is as least as possible. Thus, the following questions are raised:

1. How shall we measure the loss of information caused by a quantization ?
2. How can we find quantizations such that the loss of information is small ?

In applied data analysis many methods are suggested how to perform quantizations. It is an important point to find unifying ideas within the variety of these methods. Actually, there is such a unifying point of view which covers several common methods. In Strasser (2000b) and in Pötzlberger and Strasser (2000) the following approach to optimal quantization is considered: For any given convex function f the integral

$$\int f d\sigma^{\mathcal{B}} = \sum_{i=1}^m w_i f(t_i) = \sum_{i=1}^m \sigma(B_i) f\left(\int x \sigma(dx|B_i)\right)$$

is to be maximized where $\mathcal{B}$ varies among all partitions with at most m subsets. This construction principle covers (depending on the underlying convex function f) minimum variance partitions (resp. principal point quantizations, for $f(x) = \|x\|^2$), and Kohonen quantizations (in the sense of SOFM, for $f(x) = \|x\|$). Moreover, this principle is sufficiently general to yield a lot of new quantization methods (see Steiner (1999)) which bring about considerable advantages for practical purposes (see Strasser (2000a), as well as the references there). The number

$$I_f(\tau) := \int f d\tau$$

can be considered as a measure of information within the quantization τ . In Strasser (2000b) this interpretation is justified both heuristically and by simple decision theoretic arguments.

In the present paper we will give a well-founded analysis of this optimization principle. We will show that the classical ideas of statistical decision theory lead to this optimization principle in a straightforward manner. The basic ideas of our analysis have been applied for the first time by Bock (1996) for dichotomies (simple discrimination problems). Let us explain the arguments by Bock as an introduction into the main results of the present paper.

Bock starts with a dichotomy $E = (\Omega, \mathcal{A}, (P_1, P_2))$. The probability distributions P_1 and P_2 are to be distinguished by statistical decision functions. The statistical separability is described by a so-called f -divergence

$$I_f(E) := \int f\left(\frac{dP_2}{dP_1}\right) dP_1$$

where f is some convex function. Let us denote by $\sigma := \mathcal{L}\left(\frac{dP_2}{dP_1} \middle| P_1\right)$ the distribution of the likelihood ratio of E under P_1 . Then we have

$$I_f(E) = \int f d\sigma.$$

Bock attempts to find a partition $\mathcal{C} = (C_1, C_2, \dots, C_m)$ consisting of at most m subsets such that within the reduced experiment $E|\mathcal{C} = (\Omega, \mathcal{C}, (P_1, P_2))$ the probability measures P_1 and P_2 can be separated as well as possible. For this, the partition $\mathcal{C}$ should be such that the f -divergence of the reduced experiment

$$I_f(E|\mathcal{C}) := \int f\left(E_{P_1}\left(\frac{dP_2}{dP_1}|\mathcal{C}\right)\right) dP_1$$

is maximal.

The way to solve this problem is outlined by Bock as follows: If the distribution σ of the likelihood ratio is quantized by a partition $\mathcal{B}$ with m subset in such a way that

$$I_f(\sigma^{\mathcal{B}}) = \int f d\sigma^{\mathcal{B}}$$

is maximal, then the partition $\mathcal{C}$ defined by $C_i := \left\{\frac{dP_2}{dP_1} \in B_i\right\}$ is the solution of the original decision theoretic problem. Moreover, Bock suggests algorithms to perform the necessary quantization of σ .

These ideas by Bock show that there is a close connection between quantization of distributions and statistical decision theory. Decision theoretic problems lead to quantization problems, and conversely, quantization problems may be interpreted in a decision theoretic way. These mutual relations between statistical decision theory and quantization are the subject of this paper.

Let us give an overview over subjects and results.

In section 2 we give an introduction into the theory of majorization. This basically purely mathematical theory will be the general framework for our

view of quantization. The main theorem of majorization is a result going back to Hardy, Littlewood and Polya (1929) which relates majorization to the comparison of integrals of convex functions.

In section 3 we define our view of quantization as an optimization problem in terms of majorization. We report some fundamental results by Pötzlberger (2000a and 2000b) showing that maximizing $I_f(\tau)$ for arbitrary convex functions actually gives the complete solution of the quantization problem.

Section 4 is devoted to the relations between statistical decision theory and the theory of majorization. We explain some results of Blackwell's ((1951) and (1953)) theory of the comparison of statistical experiments. The basic result by Blackwell states that statistical experiments may be compared by the majorization order applied to the distributions of their likelihood ratios. In view of the theorem by Hardy, Littlewood and Polya (actually by some of its extensions) the comparison can be carried out in terms of integrals of convex functions. This closes the gap to the approach by Bock (1996).

In section 5 we finally consider the problem of reduction of complexity of statistical experiments. We show that looking for a most informative but finitely generated subfield is equivalent to an optimal quantization of the distribution of the likelihood ratios (of the standard measure, in terms of Blackwell's theory). By Pötzlberger's theorems it follows that quantizations which are maximal with respect to majorization can be obtained by maximizing integrals of convex functions. This supports the claim that our view of quantization is the appropriate one for statistical decision problems.

2 Majorization

The concept of majorization has a long history and a lot of applications. The range of applications starts with concentration measurement, covers a plenty of inequalities in purely mathematical areas, and extends as far as information theory, potential theory and statistical decision theory. The book by Marshall and Olkin (1979) contains a lot of information on majorization. Blackwell (1951) discovered the connection between majorization and statistical decision theory. This particular topic has been developed very far by LeCam (1964) and Torgersen (1970). These results and their extension to the general theory of majorization are presented by Torgersen (1991).

Let V be a finite dimensional vector space with Borelfield $\mathcal{F}$. Moreover let σ and τ be two probability measures on $(V, \mathcal{F})$. Throughout the following we will assume tacitely that all probability measures have finite first moments.

2.1 A special case

Suppose that the measure τ has finite support, i.e.

$$\tau = \sum_{i=1}^m w_i \delta_{t_i}.$$

Then the measure τ is said to be majorized by the measure σ if τ is built from σ by a procedure of a very particular type. We are now going to describe this type of procedure.

The transformation of σ into τ consists of two steps:

1. In the first step the measure σ is decomposed into a mixture of m probability measures. This means that we look for probability measures $\sigma_1, \dots, \sigma_m$ on $(V, \mathcal{F})$, such that σ can be written as a convex combination

$$\sigma(A) = \sum_{i=1}^m w_i \sigma_i(A), \quad A \in \mathcal{F}. \quad (1)$$

The weights $w_1, \dots, w_m$ of this convex combination will serve as weights of the resulting measure τ .

2. In the second step we compute the centroids of the measures $\sigma_1, \dots, \sigma_m$, i.e.

$$t_i = \int x \sigma_i(dx), \quad i = 1, 2, \dots, m, \quad (2)$$

and we define these centroids $t_1, \dots, t_m$ to be the mass points of the resulting measure τ .

Roughly speaking, the transformation of σ into τ can be summarized as follows: The measure τ arises from σ by desintegration and subsequent averaging.

In this way we arrive at a first definition of majorization.

Definition 1. Let σ and τ be probability measures $(V, \mathcal{F})$. The measure $\tau = \sum_{i=1}^m w_i \delta_{t_i}$ with finite support is majorized by σ , in symbols $\sigma \succ \tau$, if there are probability measures $\sigma_1, \dots, \sigma_m$ on $(V, \mathcal{F})$, such that equations (1) and (2) are valid.

In the following we will name a measure τ with finite support such that $\sigma \succ \tau$ as a quantization of σ . We are aware of the fact that this is a restriction of the common more general use of the term quantization in the scientific literature. However, we will be able to show that this restriction is appropriate for the purposes of statistical decision theory.

Clearly, a measure σ admits many desintegrations of the form (2). Therefore, in general there are many different quantizations τ of σ .

Before we turn to the general concept of majorization (where τ need not have finite support) we remind of the special case of Section 1, where the quantization $\tau = \sigma^{\mathcal{B}}$ was defined by a partition $\mathcal{B}$ of V .

2.2 The general case

In order to explain the general meaning of $\sigma \succ \tau$ we have to change the formal description of desintegration and subsequent averaging.

Let us start again from the equations (1) and (2). We define a transition probability $D : (y, A) \mapsto D(y, A)$ by

$$D(y, A) = \sigma_i(A) \quad \text{if } y = t_i, \quad i = 1, 2, \dots, m,$$

or in slightly different words

$$D(y, A) := \sum_{i=1}^m \sigma_i(A) 1_{\{t_i\}}(y), \quad A \in \mathcal{F}, y \in V.$$

This defines D for τ -almost all $y \in V$. Using D we may write the equations (1) and (2) as

$$\sigma(A) = \sum_{i=1}^m w_i \sigma_i(A) \quad \Leftrightarrow \quad \sigma(A) = \int D(y, A) \tau(dy)$$

and

$$t_i = \int x \sigma_i(dx), \quad i = 1, 2, \dots, m, \quad \Leftrightarrow \quad y = \int x D(y, dx) \quad \tau\text{-a.e.}$$

The transition probability D is called a dilation from τ to σ .

This point of view can be extended to arbitrary probability measures.

Definition 2. Let σ and τ be probability measures on V . The measure σ is a majorization of τ (in symbols $\sigma \succ \tau$), if there is a dilation from τ to σ , i.e. if there is a transition probability $D : \mathcal{F} \times V \rightarrow \mathbb{R}$ such that

$$\sigma(A) = \int D(A, y) \tau(dy), \quad A \in \mathcal{F}, \quad (3)$$

and

$$y = \int x D(dx, y) \quad \tau\text{-a.e.} \quad (4)$$

It is easy to see that for measures τ with finite support the Definitions 1 and 2 are equivalent.

2.3 Properties of majorization

The relation of majorization is a semiordering of the set of all probability measures, i.e. we have

$$\begin{aligned} \sigma &\succ \sigma, \\ \sigma &\succ \tau, \tau \succ \mu \implies \sigma &\succ \mu. \end{aligned}$$

From Definition 2 it follows that probability measures which are comparable with respect to majorization must have equal first moments.

From Jensen's inequality it follows that

$$\sigma \succ \tau \implies \int f d\sigma \geq \int f d\tau \quad (5)$$

for any convex function $f : V \rightarrow \mathbb{R}$. (Throughout the following we assume tacitely that only such convex functions admitted which are dominated from below by a affine linear function. This guarantees that all integrals are well-defined.) It follows easily that majorization is identifying, i.e.

$$\sigma \succ \tau, \tau \succ \sigma \implies \sigma = \tau.$$

The main theorem of the theory of majorization is the so-called Dilation Theorem which is also known as Theorem by Blackwell-Sherman-Stein. This theorem states that the integral inequalities in (5) are even sufficient for $\sigma \succ \tau$.

Theorem 1. (*Blackwell, Sherman, Stein*) *The following assertions are equivalent:*

1. $\sigma \succ \tau$.
2. $\int f d\sigma \geq \int f d\tau$ for all convex functions $f : \mathbb{V} \rightarrow \mathbb{R}$.

The proof of (2) $\implies$ (1) has along history, beginning with Hardy, Littlewood and Polya (1929). The multivariate version was proved by Blackwell (1951), Sherman (1951), and Stein (1951). An infinitely dimensional version for locally convex spaces was proved by V. Strassen (1965).

3 The quantization problem

Let σ be a probability measure which is to be quantized. By

$$\mathcal{M}(\sigma, m) = \{\tau : \sigma \succ \tau, |\text{supp}(\tau)| \leq m\}$$

we denote the set of all quantizations of σ supported by at most m points. Since any quantization causes some loss of information we are interested in such quantizations which are maximal with respect to majorization.

Definition 3. A quantization $\tau \in \mathcal{M}(\sigma, m)$ is called admissible if it is maximal with respect to $\succ$, i.e. if it is not majorized by any other quantization in $\mathcal{M}(\sigma, m)$.

Since majorization defines only a semiordering we cannot expect that $\mathcal{M}(\sigma, m)$ contains a maximum. But it makes sense to look for unsurpassable, i.e. admissible quantizations.

This program is seemingly different from the usual approach to look only for quantizations which are generated by partitions and which solve particular

optimization problems. Popular examples are MSL-Partitions (Bock (1996)) or MSP-Partitions (Pötzelberger and Strasser (2000)) where integrals $I_f(\tau)$ of convex functions f are maximized. Since those methods have turned out to be very useful for practical purposes it is an important point to show their distinguished role also in our general quantization framework.

Pötzelberger (2000a, 2000b) has obtained a series of results which throw some light on these problems. In some sense these results provide a complete theoretical solution. Since the proofs are mathematically rather sophisticated, in this paper we can only report some basic conclusions. We would like to emphasize that, although Pötzelberger's results are formally similar to A. Wald's fundamental theorems (complete class theorem, admissibility of Bayes solutions), they require new proofs of a completely different structure.

Suppose that σ is an absolutely continuous distribution, i.e. $\sigma \ll \lambda$ where λ denotes the Lebesgue measure. If this assumption is not satisfied then the following statements need some modifications (see Pötzelberger (2000b)).

The first result states that admissible quantizations are generated by partitions of a particular type.

Theorem 2. (Pötzelberger (2000a)) *Let $\sigma \ll \lambda$. For every admissible quantization $\mu \in \mathcal{M}(\sigma, m)$ there is a partition $\mathcal{B}$, consisting of at most m convex polytopes, such that $\mu = \sigma^{\mathcal{B}}$.*

The second result shows that we need only consider admissible quantizations.

Theorem 3. (Pötzelberger (2000a)) *Let $\sigma \ll \lambda$. For any quantization $\tau \in \mathcal{M}(\sigma, m)$ there is an admissible quantization $\mu \in \mathcal{M}(\sigma, m)$, such that $\tau \prec \mu$.*

The question arises how to construct admissible quantizations. For a convex function f let

$$\mathcal{O}_f(\sigma, m) := \left\{ \mu \in \mathcal{M}(\sigma, m) : \int f d\mu = \sup_{\tau \in \mathcal{M}(\sigma, m)} \int f d\tau \right\}.$$

The set $\mathcal{O}_f(\sigma, m)$ contains those quantizations which are obtained by optimizing integrals of convex functions. These are exactly the optimization problems which have been considered by Bock (1996), and where algorithms are provided by Bock (1996) and by Pötzelberger and Strasser (2000).

Theorem 4. (Pötzelberger (2000a)) *Let $\sigma \ll \lambda$ and let f be a convex function which is not the maximum of $m - 1$ affine linear functions. Then all quantizations in $\mathcal{O}_f(\sigma, m)$ are admissible in $\mathcal{M}(\sigma, m)$.*

This theorem implies that the optimization problem which is defined by $\mathcal{O}_f(\sigma, m)$ yields quantizations which are unsurpassable with respect to majorization.

The last result is concerned with the question whether all admissible quantizations can be obtained in this way.

Theorem 5. (Pötzelberger (2000a)) *Let $\sigma \ll \lambda$ and let $\mu \in \mathcal{M}(\sigma, m)$ be such that $|\text{supp}(\mu)| = m$. The quantization μ is admissible iff there is a sequence of convex functions f_n , none of which is a maximum of $m - 1$ affine linear functions, such that*

$$\mu = \lim_{n \rightarrow \infty} \mu_n \text{ weakly, where } \mu_n \in \mathcal{O}_{f_n}(\sigma, m).$$

The importance of the results by Pötzelberger for the theory of quantization can hardly be overestimated. Moreover, these results constitute a considerable progress in the theory of majorization.

4 Majorization and statistical decision theory

Blackwell (1951) discovered the connection between majorization and statistical decision theory. We refer to Heyer (1982), Strasser (1985), and LeCam (1986).

Let $(\Omega, \mathcal{A})$ be a measurable space and let $(P_1, P_2, \dots, P_n)$ be a vector of probability measures on $\mathcal{A}$. Then $E = (\Omega, \mathcal{A}, (P_1, P_2, \dots, P_n))$ is said to be a statistical experiment of order n . The idea of comparison of experiments is concerned with the ability of an experiment E to distinguish between the probability measures $(P_1, P_2, \dots, P_n)$ on the basis of statistical data. There are several possibilities to formalize this ability. In the theory of statistical experiments it is shown that many of these formalizations are more or less equivalent to each other. In a most simple way the comparison of experiments is described by the so-called randomization criterion.

Definition 4. An experiment $E = (\Omega_1, \mathcal{A}_1, (P_1, P_2, \dots, P_n))$ is more informative than an experiment $F = (\Omega_2, \mathcal{A}_2, (Q_1, Q_2, \dots, Q_n))$, in symbols $E \supseteq F$, if there exists a transition probability M from $(\Omega_1, \mathcal{A}_1)$ to $(\Omega_2, \mathcal{A}_2)$, such that $Q_i = MP_i$ for every $i = 1, 2, \dots, n$.

The underlying idea of this definition is that the probability measures Q_i can be built from the probability measures P_i by an additional random experiment (a so-called randomization), which is independent of the parameter i , i.e. which is the same for all P_i . Thus, the randomization does not create any further information on the parameter and therefore it cannot improve the ability to distinguish between the parameters.

The most simple case of a randomization is given by a reduction of the experiment E to a smaller field $\mathcal{C}$. This reduction is denoted by

$$E|\mathcal{C} := (\Omega, \mathcal{C}, (P_1, P_2, \dots, P_n)).$$

Obviously, we have $E \supseteq E|\mathcal{C}$.

If for two experiments E and F we have both $E \supseteq F$ and $E \subseteq F$, then the experiments are called equivalent, in symbols $E \sim F$. The equivalence $E \sim E|\mathcal{C}$ is valid iff $\mathcal{C}$ is a sufficient subfield of $\mathcal{A}$.

Experiments are represented by their standard measures. Let

$$S_n := \{x \in \mathbb{R}^n : \sum_{i=1}^n x_i = 1, x_i \geq 0 \text{ if } i = 1, 2, \dots, n\}$$

be the standard simplex in $\mathbb{R}^n$.

Definition 5. The standard measure of the experiment E is defined by

$$\sigma_E : B \mapsto \left(\frac{1}{n} \sum_k P_k \right) \left(\left(\frac{dP_i}{d \sum_k P_k} \right)_{1 \leq i \leq n} \in B \right), \quad B \in \mathcal{B}_n.$$

Standard measures are probability measures, satisfying $\sigma_E(S_n) = 1$ and

$$\int x_i \sigma_E(dx) = \frac{1}{n}, \quad 1 \leq i \leq n.$$

Every probability measure on S_n satisfying these conditions is called a standard measure.

The following theorem states how the statistical decision theory is connected to the theory of majorization.

Theorem 6. (*Blackwell (1951)*) *Let E and F be two experiments of order n . Then*

$$E \supseteq F \quad \Leftrightarrow \quad \sigma_E \succ \sigma_F.$$

This theorem implies that the comparison of experiments can be performed computationally by the comparison of their standard measures in terms of majorization. The dilation theorem yields the following corollary.

Corollary 1. *Let E and F be two experiments of order n . Then we have $E \supseteq F$ iff*

$$\int f d\sigma_E \geq \int f d\sigma_F \tag{6}$$

for all convex functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$.

Thus, the integrals of convex functions with respect to standard measures are isotonic functions with respect to the information ordering of experiments. In case of dichotomies the standard measures may be replaced by the distributions of the likelihood ratios.

Corollary 2. *Let $E = (P_1, P_2)$ and $F = (Q_1, Q_2)$ be two dichotomies. Then we have $E \supseteq F$ iff*

$$\int f \left(\frac{dP_2}{dP_1} \right) dP_1 \geq \int f \left(\frac{dQ_2}{dQ_1} \right) dQ_1 \tag{7}$$

for all convex functions $f : \mathbb{R} \rightarrow \mathbb{R}$.

Proof. Let $f : [0, \infty] \rightarrow \mathbb{R}$ be any convex function. Then it is easy to see that

$$g : (x_1, x_2) \mapsto f\left(\frac{x_1}{x_2}\right) x_2$$

is a convex function on $[0, \infty)^2$. Hence, the inequality (6) applied for g implies (7).

Let conversely be $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ be an arbitrary convex function. W.l.g. assume that $g(1, 0) = 0$. Define

$$f(t) := g\left(\frac{t}{1+t}, \frac{1}{1+t}\right)(1+t), \quad t \in [0, \infty].$$

Again it is easy to see that f is a convex function. From (7) it follows that (6) is valid for g .

The integrals in (7) are called f -divergences. The best known examples of f -divergences are the Kullback-Leibler information with $f(x) = x \log(x)$, the Hellinger-distance with $f(x) = 1 - \sqrt{x}$, and the total variation with $f(x) = (x - 1)^+$.

The f -divergences are also isotonic functions of the information semiordering of experiments and are thus interpreted as information measures.

5 Reduction of complexity and majorization

Let us turn to the problem how to obtain optimal reductions of statistical experiments to finite subfields. We will show that an optimal quantization of the standard measure will give an optimal reduction of the underlying experiment. This will provide also a theoretical justification of the approach by Bock (1996).

To begin with we recall that any finite partition generates a finite field, and that, conversely, each finite field is generated by a finite partition. Thus, we may identify finite fields and finite partitions.

Definition 6. An experiment F is of complexity $\leq m$, if there exists an equivalent experiment G , whose field is generated by a partition with at most m subsets.

Our view of a reduction of complexity of a statistical experiment E will be the following: The experiment E is to be replaced by another experiment F , which can be obtained from the original one, i.e. $E \supseteq F$, and which is of complexity $\leq m$. Given these side conditions we are looking for a reduced experiment F which is most informative. Moreover, for practical reasons, we would like to obtain a reduced experiment F of the form $F = E|C$, where C is a subfield of $\mathcal{A}$ which is generated by m subsets.

The first lemma shows how the complexity of the experiment is related to properties of the standard measure.

Lemma 1. *An experiment F is of complexity $\leq m$ iff the standard measure $\tau = \sigma_F$ is supported by at most m points.*

Proof. Let F be an experiment of complexity $\leq m$. W.l.g. let $F = (\Omega, \mathcal{C}, (Q_1, Q_2, \dots, Q_n))$, where $\mathcal{C}$ is generated by a partition $(C_1, C_2, \dots, C_m)$. Then the standard measure is given by $\sigma_F = \sum_k w_k \delta_{t_k}$, where $w_k = (\frac{1}{n} \sum_j Q_j)(C_k)$ and

$$t_k = \left(\frac{Q_i(C_k)}{\sum_j Q_j(C_k)} \right)_{1 \leq i \leq n}.$$

Conversely, let $\tau = \sum_{k=1}^m w_k \delta_{t_k}$ be a standard measure on S_n . We have to show that there exists an experiment having this standard measure and which is of complexity $\leq m$.

The solution is the so-called standard experiment of τ . We define $\Omega := \{t_1, t_2, \dots, t_m\}$, $\mathcal{A}$ to be the power set of Ω , and

$$Q_i : A \mapsto n \int_A x_i \tau(dx), \quad 1 \leq i \leq n.$$

Then it is easy to see that the experiment defined in this way has the standard measure τ .

It follows that the standard measures of experiments which are obtained from E by a reduction of complexity, are exactly the quantizations of the standard measure σ_E and therefore can be described by the set

$$\mathcal{M}(\sigma_E, m) = \{\tau : \tau \prec \sigma_E, |\text{supp}(\tau)| \leq m\}.$$

In this way we have translated the complexity reduction of experiments into the quantization of standard measures.

For dichotomies the standard measure can be transformed into the distribution of the likelihood ratio. Therefore, the complexity reduction of a dichotomy is equivalent to the quantization of the distribution of the likelihood ratio.

The next question we have to discuss is how to quantize the standard measure. It is natural to apply one of the Corollaries 1 or 2. The idea of Bock (1996) is to choose a convex function f and then to quantize the distribution of the likelihood ratio of a dichotomy by maximizing the f -divergence. For experiments of order n this corresponds to choosing a convex function f and quantizing the standard measure σ by maximizing $I_f(\sigma^B)$. We will show that this approach can be justified by principles of statistical decision theory.

Since both majorization of standard measures and information comparison of experiments are semiorderings, we cannot expect that the set $\mathcal{M}(\sigma_E, m)$ contains a maximum. However, it makes sense to look for maximal elements in $\mathcal{M}(\sigma_E, m)$.

Definition 7. An experiment $F \subseteq E$ of complexity $\leq m$ is called admissible in $\mathcal{M}(\sigma_E, m)$, if F is maximal in $\mathcal{M}(\sigma_E, m)$, i.e. if there is no other experiment G of complexity $\leq m$ such that $F \subsetneq G \subseteq E$.

From the results of Pötzelberger (2000a and 2000b) discussed in Section 3 it follows how to obtain maximal standard measures in $\mathcal{M}(\sigma_E, m)$: Choose a convex function f and maximize $I_f(\tau)$ for $\tau \in \mathcal{M}(\sigma_E, m)$.

Theorem 7. Let f be a convex function on S_n . If $\tau \in \mathcal{M}(\sigma_E, m)$ is a standard measure satisfying

$$\int f d\tau = \sup_{\mu \in \mathcal{M}(\sigma_E, m)} \int f d\mu,$$

then any experiment F whose standard measure is τ , is admissible with a complexity $\leq m$.

The results of Pötzelberger imply that in this way we may approximate all admissible complexity reductions of E .

Thus it is shown that the quantization of standard measures by maximizing integrals of convex functions constitutes the general solution of the problem of reduction of complexity for statistical experiments.

Another consequence of the results by Pötzelberger is that any admissible standard measure τ is of the form $\tau = \sigma_E^{\mathcal{B}}$ where $\mathcal{B}$ is a partition consisting of m polytopes. In Pötzelberger and Strasser (2000) algorithms are investigated which allow to compute these partitions in an constructive way. In this way we even obtain a constructive representation of the partition $\mathcal{C}$ to which the original experiment is to be reduced.

Corollary 3. Let $\tau = \sigma_E^{\mathcal{B}}$ be optimal in the sense of Theorem 7. Then the subsets

$$C_j := \left\{ \left(\frac{dP_i}{d\sum_k P_k} \right)_{1 \leq i \leq n} \in B_j \right\}, \quad j = 1, 2, \dots, m,$$

define a partition $\mathcal{C}$, such that $F := E|\mathcal{C}$ has the standard measure τ and is an admissible reduction of complexity of E of complexity $\leq m$.

A final question is how to choose the convex function f which is used for the quantization of σ_E . Our last theorem indicates that any Bayes decision problem determines a convex function which may be used for the quantization of the standard measure. Further ideas for the choice of the convex function f are contained in Strasser (2000b).

Theorem 8. Let $D = \{1, 2, \dots, r\}$ be a finite decision space, $U_i(j)$ the utility of the decision j in state i , and let $(\alpha_1, \alpha_2, \dots, \alpha_n)$ be a prior distribution.

If $U(\rho)$ denotes the Bayesian utility of a decision function ρ for that decision problem then we have

$$\sup_{\rho} U(\rho) = n \int f d\sigma_E,$$

where f denotes the convex function

$$f(x) := \max_j \sum_{i=1}^n \alpha_i U_i(j) x_i, \quad x \in \mathbb{R}^n.$$

Proof. Let $\rho_j(\omega)$ the probabilities for the randomized decision function ρ to decide in favour of j if ω was observed. Then the Bayesian utility of ρ is given by

$$\begin{aligned} R(\rho) &= \sum_{i=1}^n \sum_{j=1}^r \alpha_i U_i(j) \int \rho_j dP_i \\ &= \int \sum_{i=1}^n \sum_{j=1}^r \alpha_i U_i(j) \rho_j \frac{dP_i}{d \sum_k P_k} d \sum_k P_k. \end{aligned}$$

If we define

$$j(\omega) := \operatorname{argmax}_j \sum_{i=1}^n \alpha_i U_i(j) \frac{dP_i}{d \sum_k P_k}(\omega),$$

then the Bayes decision function is

$$\rho_j^*(\omega) := \begin{cases} 1 & \text{if } j = j(\omega), \\ 0 & \text{if } j \neq j(\omega). \end{cases}$$

For the optimal Bayesian utility it follows that

$$\begin{aligned} R(\rho^*) &= \int \max_j \left(\sum_{i=1}^n \alpha_i U_i(j) \frac{dP_i}{d \sum_k P_k} \right) d \sum_k P_k \\ &= n \int \max_j \left(\sum_{i=1}^n \alpha_i U_i(j) x_i \right) \sigma_E(dx) \\ &= n \int f d\sigma_E. \end{aligned}$$

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